A proof of the conjectured generating function for OEIS A266072

Adrian Perez Fontelles
Independent researcher

31 August 2026

Abstract

OEIS A266072 carries a conjectured generating function. We prove it. The entry states, as fact, a closed form, and from that a linear recurrence with polynomial coefficients satisfied by the sequence is derived exactly. The conjectured generating function is then shown to satisfy the same recurrence, and to agree with the sequence at enough consecutive indices. A recurrence of order 3 whose leading coefficient does not vanish, together with 3 consecutive values, determines a sequence completely, so the two agree everywhere. Nothing is checked numerically and then assumed: the recurrence is derived symbolically, the verification is an exact identity, and the finitely many indices where the leading coefficient vanishes are computed rather than waved at.

2020 Mathematics Subject Classification. 05A15, 11B37, 33F10.

1 The sequence, what is known, and what is conjectured

OEIS A266072 is “Number of ON (black) cells in the n-th iteration of the "Rule 3" elementary cellular automaton starting with a single ON (black) cell”. It has offset 0 and begins

$$a(0), \ldots, a(7) \ = \ 1, 1, 1, 5, 1, 9, 1, 13, \ \ldots$$

The entry states, not as a conjecture,

a(n) = n-(-1)^n*(n-1).

and separately carries the comment

Conjectured g.f.: (1 + x - x^2 + 3*x^3)/(-1 + x^2)^2. - *Michael De Vlieger*, Dec 21 2015

As of the “Last modified” line on the live entry (2025-02-16, revision 30) the second statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

2 A recurrence from what is known

The closed form the entry posts is a sum of finitely many hypergeometric terms. For terms $t_1, \ldots, t_m$ one may seek $c_0, \ldots, c_m$ in $\mathbb{Q}(n)$ with $\sum_k c_k(n)t_j(n+k) = 0$ for every j; dividing the j-th equation by $t_j(n)$ makes every coefficient a rational function, so this is m equations in $m+1$ unknowns over $\mathbb{Q}(n)$ and a nonzero solution exists. It gives

$$(-n-2)\ a(n) + (-n-3)\ a(n+1) + n\, a(n+2) + (n+1)\ a(n+3) \ = \ 0 \qquad (1)$$

The leading coefficient of (1) is $n+1$. Its only integer zeros are $n \in \{-1\}$, all below the range claimed here.

3 The conjectured description satisfies it

Write $\theta = x\,\frac{d}{dx}$ and, for the coefficients p_i of (1) written in the backward form $\sum_i p_i(n)a(n-i) = 0$, set $B(x) = \sum_i x^i\,(p_i(\theta+i)A)(x)$ for the conjectured generating function A. Then $[x^n]B = \sum_i p_i(n)a(n-i)$, so the conjectured generating function has coefficients satisfying (1) exactly when B is a polynomial. Reducing B inside the field A generates, which is closed under θ, gives a polynomial, and the verification is an exact cancellation of rational functions.

4 Uniqueness

Lemma 1. *Let $\sum_{j=0}^{r} q_j(n)\,b(n+j) = 0$ be a linear recurrence whose leading coefficient q_r has no zero at any integer $n \ge n_0$. If two sequences both satisfy it for all $n \ge n_0$ and agree at the r consecutive indices $n_0, \dots, n_0+r-1$, they agree at every $n \ge n_0$.*

Proof. Induction. Suppose the two agree at $n_0, \dots, m+r-1$ for some $m \ge n_0$. The recurrence at $n = m$ determines $b(m+r)$ from the preceding r values, because $q_r(m)$ is nonzero and may be divided by; both sequences therefore take the same value at $m+r$. □

Theorem 2. *For all $n \ge 0$,*

$$\sum_{n\ge 0} a(n)x^n \;=\; \frac{3x^3 - x^2 + x + 1}{(x^2-1)^2}$$

Proof. Both sides satisfy (1): the sequence because the recurrence was derived from the entry's own a closed form, and the conjectured generating function by the computation of the previous section. They agree at the 3 consecutive indices beginning $n = 0$, which was checked against the entry's DATA in exact integer arithmetic. The leading coefficient has no zero at any integer $n \ge 0$, so Lemma 1 applies. □

5 Verification

Three checks, independent of one another.

First, the description the entry states as fact was itself confirmed against the entry's DATA before any recurrence was derived from it; a statement that did not reproduce the entry's own terms would not have been used as the basis of anything.

Second, the derived recurrence (1) was evaluated on the published terms in exact integer arithmetic and vanishes at every index where it applies.

Third, the conjectured generating function was evaluated at the first 13 indices and compared term by term with the DATA; the values agree exactly. This is not part of the proof, which is symbolic, but it would catch a transcription error in either statement.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A266072.

[2] M. Petkovšek, H. S. Wilf and D. Zeilberger, *A=B*, A K Peters, 1996.

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011.